Special Relativity Notes

Charge and Current Density of a Wire in Two Frames, and the Transformation of Force

3/11/26

Velocities and symbols used throughout (please read first)

- S — the frame in which the wire is at rest (the positive charges are at rest in S).
- S' — a frame moving along the $+x$ axis with velocity $\vec{v}$ **relative to** S (a boost along the x-axis). $v = |\vec{v}|$.
- $\vec{u}$ — the **drift velocity of the electrons** in S (along the x-axis), with u_x its x-component. In the force section, $\vec{u}$ is instead the **velocity of the test charge** in S (the board writes it as U or u).
- $\beta = v/c, \qquad \gamma = \dfrac{1}{\sqrt{1 - v^2/c^2}}$ (c = speed of light). These always refer to the frame velocity v, never to u.
- Unprimed quantities are measured in S; primed quantities in S'.
- **Note:** in the previous set of notes (SR_18) the frame speed was called u and the carrier speed v; here the roles are *swapped*: frame speed = v, carrier/test-charge speed = u.

1 Recap: the Four-Current and its Lorentz Transformation

Charge density ρ and current density $\vec{J}$ combine into one four-vector:

$$J^\mu = \left(J^0,\ \vec{J}\right), \qquad J^0 = \rho c$$

What each term means

- J^μ — the **four-current**, a Lorentz four-vector with components $\mu = 0, 1, 2, 3$.
- $J^0 = \rho c$ — the **time component**. ρ is the charge per unit volume ($\mathrm{C/m^3}$); multiplying by c gives it the same units as $\vec{J}$ ($\mathrm{A/m^2}$), just as $x^0 = ct$ has the units of length.
- $\vec{J} = (J_x, J_y, J_z)$ — the **current density** (current per unit area); the spatial part, like $\vec{x}$ in $(ct, \vec{x})$.

For a boost along the x-axis with velocity v (the same rule as for (ct, x) or $(\mathcal{E}/c, p_x)$):

$$J'_x = \gamma\left(J_x - \beta J^0\right) = \frac{J_x - v\rho}{\sqrt{1 - \dfrac{v^2}{c^2}}}$$

$$J'_y = J_y$$

$$J'_z = J_z$$

$$J'^0 = \gamma\left(J^0 - \beta J_x\right)$$

Since $J'^0 = \rho' c$, dividing the last line by c gives

$$\rho' c = \gamma\left(\rho c - \frac{v}{c} J_x\right) \quad \Longrightarrow \quad \rho' = \frac{\rho - \dfrac{v}{c^2} J_x}{\sqrt{1 - \dfrac{v^2}{c^2}}}$$

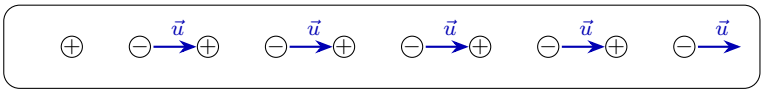
Result: transformation of ρ and $\vec{J}$ (boost along x)

$$J'_x = \frac{J_x - v\rho}{\sqrt{1 - \dfrac{v^2}{c^2}}}, \qquad J'_y = J_y, \qquad J'_z = J_z, \qquad \rho' = \frac{\rho - \dfrac{v}{c^2} J_x}{\sqrt{1 - \dfrac{v^2}{c^2}}}$$

where: ρ, $\vec{J}$ are measured in S; ρ', $\vec{J}\,'$ in S'; v is the velocity of S' relative to S (along x); $\beta = v/c$; $\gamma = 1/\sqrt{1 - v^2/c^2}$.

Only the component **along** the boost (J_x) mixes with ρ; the components perpendicular to the boost are unchanged.

2 A Current-Carrying Wire: Charge and Current Density



positive ions: at rest in S

electrons: drift velocity $\vec{u}$ (along $+x$)

charge density: $\rho = \dfrac{\rho_0}{\sqrt{1 - \dfrac{u^2}{c^2}}}$ current density: $\vec{J} = \rho\,\vec{u}, \quad J_x = \rho\, u_x$

where: ρ_0 is the charge density of the moving carriers measured in their own rest frame; ρ is the charge density in the frame where they move with speed u (their volume is length-contracted by $\sqrt{1 - u^2/c^2}$, so the density rises by the same factor); $\vec{u}$ is the drift velocity; u_x is its x-component.

3 The Wire as Seen from Frame S (wire at rest)

In S: the **positive charges are at rest** and the **electrons move with drift velocity** $\vec{u}$ (along the x-axis).

Let N be the number of electrons per unit volume, and also the number of positive charges per unit volume (the wire is neutral), and e the magnitude of the electron charge.

Charge density:

$$\rho = \rho^- + \rho^+ = -Ne + Ne = 0 \qquad \text{(net charge)}$$

Current density: only the electrons carry current (the positive charges are at rest, so $J_x^+ = 0$):

$$\vec{J} = J_x^- = \rho^- u_x = -Ne\, u_x$$

where: $\rho^- = -Ne$ is the charge density of the electrons, $\rho^+ = +Ne$ that of the positive ions, $J_x^{\pm}$ their x-current densities, u_x the electron drift velocity.

In S there is a current, so a **magnetic field exists around the wire**. But the net charge density is zero ($\rho_{\text{net}} = 0$), so there is **no electric field**.

4 The Same Wire as Seen from Frame S'

Consider an observer in S' (moving with velocity $\vec{v}$ relative to S), describing the same situation in his own language. Apply the transformation of Section 1 separately to the electrons and to the positive charges.

Charge densities in S'

Electrons ($\rho^- = -Ne$, $J_x^- = -Ne\,u_x$):

$$\rho^{-\prime} = \frac{\rho^- - \dfrac{v}{c^2}J_x^-}{\sqrt{1-\dfrac{v^2}{c^2}}} = \frac{-Ne - \dfrac{v}{c^2}(-Ne)\,u_x}{\sqrt{1-\dfrac{v^2}{c^2}}} = \frac{-Ne\left(1-\dfrac{v\,u_x}{c^2}\right)}{\sqrt{1-\dfrac{v^2}{c^2}}}$$

Positive charges ($\rho^+ = Ne$, $J_x^+ = 0$):

$$\rho^{+\prime} = \frac{\rho^+ - \dfrac{v}{c^2}J_x^+}{\sqrt{1-\dfrac{v^2}{c^2}}} = \frac{Ne}{\sqrt{1-\dfrac{v^2}{c^2}}}$$

Total:

$$\rho' = \rho^{-\prime} + \rho^{+\prime} = \frac{Ne}{\sqrt{1-\dfrac{v^2}{c^2}}} - \frac{Ne\left(1-\dfrac{v\,u_x}{c^2}\right)}{\sqrt{1-\dfrac{v^2}{c^2}}}$$

$$\boxed{\rho' = \frac{Ne\,\dfrac{v\,u_x}{c^2}}{\sqrt{1-\dfrac{v^2}{c^2}}} \neq 0} \quad \Longrightarrow \quad \textbf{an electric field exists in } S'.$$

where: N = number density of each species in S; e = electron charge magnitude; v = velocity of S' relative to S; u_x = electron drift velocity in S; ρ' = net charge density seen in S'.

The wire that is electrically neutral in S is therefore *charged* in S'.

Current density in S'

$$J_x' = J_x^{-\prime} + J_x^{+\prime}$$

Positive charges:

$$J_x^{+\prime} = \frac{J_x^+ - v\rho^+}{\sqrt{1-\dfrac{v^2}{c^2}}} = \frac{-vNe}{\sqrt{1-\dfrac{v^2}{c^2}}}$$

Electrons:

$$J_x^{-\prime} = \frac{J_x^- - v\rho^-}{\sqrt{1-\dfrac{v^2}{c^2}}} = \frac{-Ne\,u_x + vNe}{\sqrt{1-\dfrac{v^2}{c^2}}} = \frac{Ne\,(v-u_x)}{\sqrt{1-\dfrac{v^2}{c^2}}}$$

Total:

$$\boxed{J'_x = J_x^{-\prime} + J_x^{+\prime} = \frac{-Ne\,u_x}{\sqrt{1-\dfrac{v^2}{c^2}}}} \quad \Longrightarrow \quad \textbf{a magnetic field exists in } S'.$$

where: $J_x^{\pm\prime}$ are the x-current densities of the positive/negative charges in S', and the other symbols are as above.

Consistency check with velocities. In S' the positive charges (at rest in S) move with velocity $-v$, so $J_x^{+\prime} = \rho^{+\prime}(-v) = -vNe/\sqrt{1-v^2/c^2}$, matching the result above. The electrons move with $u'_x = \dfrac{u_x - v}{1 - vu_x/c^2}$ (relativistic velocity addition), and $\rho^{-\prime}u'_x = \dfrac{Ne\,(v-u_x)}{\sqrt{1-v^2/c^2}}$, again matching.

Conclusion

One physical situation, two descriptions: in S there is only a magnetic field; in S' there is *both* an electric and a magnetic field. So $\vec{E}$ and $\vec{B}$ are not independent — they mix under a Lorentz transformation.

5 How Are $\vec{E}$ and $\vec{B}$ Related? Transformation of Force

Lorentz force on a test charge q_t in frame S:

$$\vec{F} = q_t\left(\vec{E} + \vec{u}\times\vec{B}\right)$$

where: q_t = test charge; $\vec{E}, \vec{B}$ = fields in S; $\vec{u}$ = velocity of the test charge in S.

Question: what is $\vec{F}\,'$ in the frame S'?

Force is the rate of change of momentum, $\vec{F} = \dfrac{d\vec{p}}{dt}$. In S the components are (F_x, F_y, F_z) with $F_x = \dfrac{dp_x}{dt}$. In S':

$$F'_x = \frac{dp'_x}{dt'} = \frac{dp'_x/dt}{dt'/dt}$$

Numerator: dp'_x/dt

Momentum and energy transform like (x, t):

$$p'_x = \frac{p_x - \dfrac{v}{c^2}\mathcal{E}}{\sqrt{1-\dfrac{v^2}{c^2}}}$$

where: p_x = x-momentum of the test charge in S; $\mathcal{E}$ = its energy in S; v = velocity of the (relativistic) frame S'.

Differentiate with respect to t (the time of S):

$$\frac{dp'_x}{dt} = \frac{\dfrac{dp_x}{dt} - \dfrac{v}{c^2}\dfrac{d\mathcal{E}}{dt}}{\sqrt{1-\dfrac{v^2}{c^2}}} = \frac{F_x - \dfrac{v}{c^2}\,(\vec{F}\cdot\vec{u})}{\sqrt{1-\dfrac{v^2}{c^2}}}$$

using $\dfrac{dp_x}{dt} = F_x$ and $\dfrac{d\mathcal{E}}{dt} = \vec{F}\cdot\vec{u}$ (proved in Section 6).

Denominator: dt'/dt

From $t' = \gamma\left(t - \dfrac{v\,x}{c^2}\right)$ and $\dfrac{dx}{dt} = u_x$ (the test charge's velocity along x):

$$\frac{dt'}{dt} = \gamma\left(1 - \frac{v\,u_x}{c^2}\right) = \frac{1-\dfrac{\beta u_x}{c}}{\sqrt{1-\dfrac{v^2}{c^2}}}$$

Result

Dividing, the factors $\gamma = 1/\sqrt{1-\dfrac{v^2}{c^2}}$ cancel in F'_x:

Transformation of force (boost along x)

$$\boxed{F'_x = \frac{F_x - \dfrac{\beta}{c}\,(\vec{F}\cdot\vec{u})}{1-\dfrac{\beta\,u_x}{c}}}$$

$$\boxed{F'_y = \frac{F_y}{\gamma\left(1-\dfrac{\beta\,u_x}{c}\right)}} \qquad \boxed{F'_z = \frac{F_z}{\gamma\left(1-\dfrac{\beta\,u_x}{c}\right)}}$$

where: $\vec{F} = (F_x, F_y, F_z)$ is the force in S and $\vec{F}\,'$ the force in S'; $\vec{u}$ is the velocity of the test charge in S (with u_x its x-component); $\vec{F}\cdot\vec{u}$ is the power delivered to the charge (rate of change of its energy); $\beta = v/c$ with v the velocity of S' relative to S; $\gamma = 1/\sqrt{1-v^2/c^2}$.

(For the transverse components, $p'_y = p_y$ and $p'_z = p_z$, so $dp'_y/dt = F_y$, $dp'_z/dt = F_z$; dividing by dt'/dt gives the F'_y, F'_z above.)

6 Exercise: Proof that $\dfrac{d\mathcal{E}}{dt} = \vec{F}\cdot\vec{u}$

For a particle of rest mass m_0 moving with velocity $\vec{u}$ in S:

$$\mathcal{E} = \frac{m_0 c^2}{\sqrt{1-\dfrac{u^2}{c^2}}}, \qquad \vec{p} = \frac{m_0\vec{u}}{\sqrt{1-\dfrac{u^2}{c^2}}}$$

where: m_0 = rest mass; $\vec{u}$ = velocity of the particle (so $u^2 = \vec{u}\cdot\vec{u}$); $\mathcal{E}$ = total energy; $\vec{p}$ = momentum.

Rate of change of energy

$$\frac{d\mathcal{E}}{dt} = m_0 c^2 \frac{d}{dt}\left(1 - \frac{u^2}{c^2}\right)^{-1/2} = m_0 c^2 \left(-\frac{1}{2}\right)\left(1 - \frac{u^2}{c^2}\right)^{-3/2}\left(-\frac{2}{c^2}\,\vec{u}\cdot\frac{d\vec{u}}{dt}\right)$$
$$= \frac{m_0}{\left(1 - \dfrac{u^2}{c^2}\right)^{3/2}}\,\vec{u}\cdot\frac{d\vec{u}}{dt}$$

Rate of change of momentum

$$\frac{d\vec{p}}{dt} = \frac{m_0}{\sqrt{1 - \dfrac{u^2}{c^2}}}\frac{d\vec{u}}{dt} + m_0\vec{u}\,\frac{d}{dt}\left(1 - \frac{u^2}{c^2}\right)^{-1/2} = \frac{m_0}{\sqrt{1 - \dfrac{u^2}{c^2}}}\frac{d\vec{u}}{dt} + \frac{m_0\vec{u}}{c^2\left(1 - \dfrac{u^2}{c^2}\right)^{3/2}}\left(\vec{u}\cdot\frac{d\vec{u}}{dt}\right)$$

Dot with $\vec{u}$:

$$\frac{d\vec{p}}{dt}\cdot\vec{u} = \frac{m_0}{\sqrt{1 - \dfrac{u^2}{c^2}}}\left(\vec{u}\cdot\frac{d\vec{u}}{dt}\right) + \frac{m_0 u^2}{c^2\left(1 - \dfrac{u^2}{c^2}\right)^{3/2}}\left(\vec{u}\cdot\frac{d\vec{u}}{dt}\right)$$
$$= \frac{m_0}{\sqrt{1 - \dfrac{u^2}{c^2}}}\left(\vec{u}\cdot\frac{d\vec{u}}{dt}\right)\left[1 + \frac{u^2/c^2}{1 - u^2/c^2}\right] = \frac{m_0}{\sqrt{1 - \dfrac{u^2}{c^2}}}\left(\vec{u}\cdot\frac{d\vec{u}}{dt}\right)\frac{1}{1 - u^2/c^2}$$
$$= \frac{m_0}{\left(1 - \dfrac{u^2}{c^2}\right)^{3/2}}\,\vec{u}\cdot\frac{d\vec{u}}{dt}$$

The two results are identical, so

$$\therefore \quad \frac{d\mathcal{E}}{dt} = \frac{d\vec{p}}{dt}\cdot\vec{u} \qquad \Longrightarrow \qquad \boxed{\frac{d\mathcal{E}}{dt} = \vec{F}\cdot\vec{u}}$$

where: $\vec{F} = d\vec{p}/dt$ is the force, $\vec{u}$ the particle velocity, and $\vec{F}\cdot\vec{u}$ the power delivered to the particle.